



Capstone Project 2

CMU-SE 451

User Story

Version 1.0

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Sign Language Detection System

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Table of Contents

1. Project Overview.....	3
2. Introduction.....	4
2.1. Purpose.....	4
2.2. Scope.....	4
2.3. References.....	4
3. Product Backlog.....	5
3.1. User Authentication and Account Management.....	5
3.1.1. User Story 1.....	5
3.1.2. User Story 2.....	5
3.1.3. User Story 3.....	5
3.1.4. User Story 4.....	6
3.1.5. User Story 5.....	6
3.1.6. User Story 6.....	7
3.2. Communication and Messaging.....	7
3.2.1. User Story 7.....	7
3.2.2. User Story 8.....	7
3.2.3. User Story 9.....	8
3.2.4. User Story 10.....	8
3.2.5. User Story 11.....	8
3.3. Sign Language Recognition.....	9
3.3.1. User Story 12.....	9
3.3.2. User Story 13.....	9
3.3.3. User Story 14.....	10
3.3.4. User Story 15.....	10
3.3.5. User Story 16.....	10
3.3.6. User Story 17.....	11
3.4. Video Calls and Conferencing.....	11
3.4.1. User Story 18.....	11
3.4.2. User Story 19.....	12
3.4.3. User Story 20.....	12
3.4.4. User Story 21.....	13
3.5. User Management and Administration.....	13
3.5.1. User Story 22.....	13
3.5.2. User Story 23.....	13
4. Approval.....	15

Revision history

Version	Date	Description of Changes	Author
1.0	15/02/2024	Initial document	Team

1. Project Overview**PROJECT INFORMATION**

Project acronym	SLDS		
Project Title	Sign Language Detection System		
Start Date	15 - Feb - 2024	End Date	20 - May - 2024
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2. Introduction

In software development, a User Story is one or more sentences that describe the natural language (not the technical language) or the business language of the end user or system user to record. What users need to do or not do their job or duty. User stories are used in methods such as Agile software development as the basis for defining system functions such as business development and to facilitate management of requirements. Help yourself know “who”, “what” and “why” in a short, simple request, often limited only by details that can be written on a piece of sticky note or scrap paper small.

User stories written by business users are the primary way of users because they understand most of the system's processes, workflows, and functions so it is working right now. User stories can also be written by Dev to make non-functional requirements (like security, performance, quality, etc.).

2.1. Purpose

- Provide a prioritized features list, containing a short description of all functionality desired in the product.
- Lists everything that the product owner and Scrum team feels should be included in the software they are developing.

2.2. Scope

- Write all the user's requirements.
- Lists some main functions of the system and acceptance criteria of each function.
- Short description of all the functionality desired in the product.
- Given the priority of each feature and function of the product.

2.3. References

No	Source	Note
1	https://www.visual-paradigm.com/guide/agile-software-development/what-is-user-story/	What is User Story?
2	https://www.atlassian.com/agile/project-management/user-stories	User Stories with Examples and Template

3. Product Backlog

3.1. User Authentication and Account Management

3.1.1. User Story 1

As a user, I want to sign up for the system

- Acceptance Criteria:
 - Users should be directed to a sign-up page upon accessing the system for the first time.
 - The sign-up page should include fields for entering necessary information such as name, email, and password.
 - Upon submission, the system should validate the entered information for completeness and accuracy.
- Severity: Medium
- Estimation Points: 5

3.1.2. User Story 2

As a user, I want to sign in to the system.

- Acceptance Criteria
 - Users should be presented with a sign-in form upon accessing the system.
 - The sign-in form should include fields for entering email and password.
 - Upon submission, the system should validate the provided credentials.
 - If the credentials are valid, the user should be authenticated and redirected to their dashboard.
 - Invalid login attempts should trigger appropriate error messages to notify the user.
- Severity: Medium
- Estimation Points: 3

3.1.3. User Story 3

As a user, I want to create a new user account

- Acceptance Criteria
 - Users should have the option to create a new user account from their profile settings.

- The user account creation form should include fields for updating personal information such as name, email, and profile picture.
- Upon submission, the system should validate the entered information and update the user's account details.
- Users should receive a confirmation message upon successful account creation.
- Severity: Medium
- Estimation Points: 5

3.1.4. User Story 4

As an admin, I want to edit user accounts

- Acceptance Criteria
 - Admin users should have access to a user management interface.
 - The user management interface should display a list of all registered users.
 - Admins should be able to select a user from the list to edit their account details.
 - Upon selecting a user, admins should be presented with a form to update the user's information.
 - Changes made by admins should be saved and reflected immediately in the system.
- Severity: High
- Estimation Points: 8

3.1.5. User Story 5

As an admin, I want to remove user accounts

- Acceptance Criteria
 - Admin users should have access to a user management interface.
 - The user management interface should display a list of all registered users.
 - Admins should be able to select one or more users from the list to remove their accounts.
 - Upon confirmation, the selected user accounts should be deactivated and removed from the system.
 - Removed users should no longer have access to the platform or its features.

- Severity: High
- Estimation Points: 8

3.1.6. User Story 6

As a user, I want to view detailed user profiles

- Acceptance Criteria
 - Users should have access to a user profile page displaying their own information.
 - The user profile page should include details such as name, email, profile picture, and any additional information provided during registration.
 - Users should be able to view detailed profiles of other users within the platform.
 - Profile views should respect user privacy settings and permissions.
- Severity: Medium
- Estimation Points: 5

3.2. Communication and Messaging

3.2.1. User Story 7

As a user, I want to add messages to a conversation

- Acceptance Criteria
 - Users should have access to a text input field within a conversation interface.
 - Users should be able to type messages and send them to the conversation.
 - Sent messages should be displayed in real-time to all participants of the conversation.
- Severity: High
- Estimation Points: 8

3.2.2. User Story 8

As a user, I want real-time messaging capabilities

- Acceptance Criteria
 - The system should provide real-time updates for new messages within conversations.
 - Messages should be delivered and displayed to all participants with minimal latency.

- Severity: High
- Estimation Points: 8

3.2.3. User Story 9

As a user, I want to delete messages in a conversation

- Acceptance Criteria
 - Users should have access to a list of all conversations they are currently participating in.
 - The conversation list should display relevant details such as the names of other participants and the most recent message.
 - Users should be able to click on a conversation to view its full details and history.
 - The conversation list should support filtering and searching for specific conversations.
- Severity: Medium
- Estimation Points: 5

3.2.4. User Story 10

As a user, I want to list all conversations I'm a part of

- Acceptance Criteria
 - Users should have access to a list of all conversations they are currently participating in.
 - The conversation list should display relevant details such as the names of other participants and the most recent message.
 - Users should be able to click on a conversation to view its full details and history.
 - The conversation list should support filtering and searching for specific conversations.
- Severity: Medium
- Estimation Points: 5

3.2.5. User Story 11

As a user, I want to list all messages within a conversation

- Acceptance Criteria

- Users should have access to view all messages within a conversation.
- The message list should display messages in chronological order, with the newest messages at the bottom.
- Each message should include details such as the sender's name, timestamp, and message content.
- The message list should support scrolling and loading older messages as needed.
- Severity: Medium
- Estimation Points: 5

3.3. Sign Language Recognition

3.3.1. User Story 12

As a user, I want to train the model for sign language recognition

- Acceptance Criteria
 - Users should have access to a training interface or tool within the system.
 - The training tool should allow users to upload and label sign language data for model training.
 - The system should provide options to customize and optimize the training parameters.
 - Upon completion, the trained model should be integrated into the sign language recognition system.
- Severity: High
- Estimation Points: 8

3.3.2. User Story 13

As a user, I want to convert text to sign language

- Acceptance Criteria
 - Users should have access to a text input field for entering the desired text.
 - Upon submission, the system should process the text and generate corresponding sign language gestures.
 - The generated sign language should accurately represent the input text and follow standard sign language conventions.

- Users should have the option to select different sign language dialects or styles.
- Severity: Medium
- Estimation Points: 5

3.3.3. User Story 14

As a user, I want to convert voice to sign language

- Acceptance Criteria
 - Users should have access to a voice input feature for speaking phrases or sentences.
 - The system should process the voice input and transcribe it into text.
 - Upon transcription, the system should convert the text into corresponding sign language gestures.
 - The sign language interpretation should accurately represent the spoken content.
- Severity: High
- Estimation Points: 8

3.3.4. User Story 15

As a user, I want to convert sign language to text and voice

- Acceptance Criteria
 - Users should have access to a sign language input feature for performing gestures.
 - The system should recognize and interpret the sign language gestures, converting them into text.
 - Upon interpretation, the system should provide an option to convert the recognized text into spoken language.
 - The spoken language interpretation should be audible to the user.
- Severity: High
- Estimation Points: 8

3.3.5. User Story 16

As a user, I want to convert voice to sign language during video calls

- Acceptance Criteria
 - Users participating in a video call should have access to a voice input feature.

- The system should process the voice input and convert it into corresponding sign language gestures.
- The sign language gestures should be displayed in real-time to all participants during the video call.
- Users should have the option to enable or disable this feature during the call.
- Severity: High
- Estimation Points: 8

3.3.6. User Story 17

As a user, I want to convert sign language to voice during video calls

- Acceptance Criteria
 - Users participating in a video call should have access to a sign language input feature.
 - The system should recognize and interpret the sign language gestures, converting them into text.
 - Upon interpretation, the system should convert the recognized text into spoken language.
 - The spoken language interpretation should be audible to all participants during the video call.
- Severity: High
- Estimation Points: 8

3.4. Video Calls and Conferencing

3.4.1. User Story 18

As a user, I want to create a room for video calls

- Acceptance Criteria
 - Users should have access to a feature to create a new room for video calls.
 - Upon creation, the system should generate a unique room ID and provide a link to share with other participants.
 - The room creator should have the ability to customize room settings, such as enabling/disabling the camera or microphone.

- Participants should be able to join the room using the provided link without requiring an account.
- Severity: Medium
- Estimation Points: 5

3.4.2. User Story 19

As a user, I want to leave a room during a video call

- Acceptance Criteria
 - Users participating in a video call should have access to a "Leave Room" button or option.
 - Clicking on the "Leave Room" option should immediately disconnect the user from the ongoing video call.
 - Users should no longer have access to the room's audio or video feeds upon leaving.
 - The system should handle the disconnection smoothly without disrupting other participants.
- Severity: Medium
- Estimation Points: 3

3.4.3. User Story 20

As a user, I want to search for other users within the platform

- Acceptance Criteria
 - Users should have access to a search feature to find other registered users within the platform.
 - The search functionality should support searching by user name, email, or other relevant criteria.
 - Search results should be displayed in a clear and organized manner, with relevant user information.
 - Users should be able to click on a search result to view the detailed user profile or initiate a conversation.
- Severity: Medium

- Estimation Points: 5

3.4.4. User Story 21

As a user, I want to add other users to a conversation.

- Acceptance Criteria
 - Users should have access to an "Add User" or "Invite User" button within the conversation interface.
 - Upon clicking the button, users should be prompted to enter the username or email of the user they want to add.
 - The system should send an invitation to the selected user to join the conversation.
 - Upon accepting the invitation, the added user should be able to view and participate in the conversation.
- Severity: Medium
- Estimation Points: 5

3.5. User Management and Administration

3.5.1. User Story 22

As a user, I want to send email invitations to other user

- Acceptance Criteria
 - Users should have access to an "Invite User" button or interface within the platform.
 - Upon clicking the "Invite Users" button, users should be prompted to enter the email addresses of the users they want to invite.
 - The system should send email invitations to the specified email addresses with a personalized message and a registration link.
 - Users should receive a confirmation message upon successful email delivery.
- Severity: Medium
- Estimation Points: 5

3.5.2. User Story 23

As a user, I want to list all users registered on the platform

- Acceptance Criteria

- Users should have access to a "List Users" feature within the platform.
 - The user directory should display a list of all registered users on the platform.
 - Each user entry in the directory should include details such as username, email, and profile picture.
 - Users should be able to click on a user entry to view the detailed profile of the user.
- Severity: Medium
 - Estimation Points: 5

4. Approval